Renewable Energy (Solar/Wind) Output Fluctuation Analysis

Project Document

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Abstract

This project analyzes how the power output of solar and wind generation fluctuates over time and identifies the weather conditions behind those fluctuations. Using time-series analysis, fluctuation metrics and baseline forecasting models, it aims to support better grid balancing, storage planning and energy scheduling.

1. Problem Statement

- Solar and wind generation depend on weather, so their output rises and falls minute by minute and season by season.
- These swings make grid balancing, storage sizing and energy trading harder and more costly.
- Goal: measure, explain and forecast the fluctuation of solar and wind output using data.

2. Objectives

- Quantify variability of solar and wind output (hourly, daily, seasonal).
- Identify the weather drivers: irradiance, cloud cover, temperature, wind speed.
- Compare solar vs. wind behaviour and how they complement each other.
- Build a baseline forecasting model and evaluate its accuracy.

3. Data & Methodology

- Data: time-series of power output with weather variables (public energy and weather datasets).
- Cleaning: handle missing values, outliers, timestamp alignment, unit consistency.
- EDA: trends, seasonality, daily profiles, correlation with weather features.
- Modelling: baseline (persistence) vs. regression / tree-based / time-series models.

4. Fluctuation Metrics

- Coefficient of variation and standard deviation of output.

- Ramp rate: change in output between consecutive intervals.
- Capacity factor and its variation by hour, day and month.
- Autocorrelation and spectral view of daily / seasonal cycles.

5. Key Insights (Expected)

- Solar follows a strong daily cycle with sharp ramps at sunrise, sunset and cloud events.
- Wind is less predictable hour to hour but stronger in specific seasons and at night.
- Combining solar and wind smooths total output compared with either alone.
- Weather features explain a large share of the variance in output.

6. Tools & Technologies

- Python: pandas, NumPy for processing; matplotlib / seaborn for visualization.
- scikit-learn and statsmodels for modelling and evaluation.
- Jupyter Notebook for analysis; GitHub for version control and delivery.

7. Expected Deliverables

- Cleaned dataset and reproducible analysis notebook.
- Visual dashboard of fluctuation patterns and ramp events.
- Forecasting model with error metrics (MAE, RMSE).
- Project document, presentation and demo video.

8. Conclusion

- Understanding output fluctuation is key to a reliable, renewable-powered grid.
- Data analysis and forecasting reduce balancing cost and support storage planning.
- Next steps: extend to more sites, finer time resolution and advanced models.